

MIKE WONG

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SUMMARY

Fifth-year CS PhD candidate at Princeton University building AI systems for efficient, real-world inference and deployment. My work develops application-aware systems that identify and process only task-relevant information, reducing AI cost, latency, and resource consumption across LLM agents, vision models, and distributed infrastructure. At Princeton and Microsoft Research, I have led systems spanning speculative execution for web agents, adaptive vision, SmartNIC-accelerated AI pipelines, and production-scale network telemetry, with publications at EuroSys and NSDI and additional work at SIGCOMM and CoNEXT. Research from my Microsoft collaboration is being explored for deployment in Microsoft's production infrastructure.

EDUCATION

Princeton University *Aug 2021 – Late 2026 (expected)*
Ph.D. in Computer Science. Advisor: Ravi Netravali. M.A. conferred Sep 2023.

New York University *Aug 2015 – Dec 2019*
B.A. in Computer Science, minor in Mathematics. Graduated with Honors.

EXPERIENCE

Graduate Research Assistant, Princeton University *Aug 2021 – Present*
Advisor: Ravi Netravali

- Built an adaptive camera-server system that dynamically filters and prioritizes spatial video data, reducing live vision inference cost by **2–3.7×** while matching or improving tracking accuracy (MadEye, NSDI '24, first author).
- Designed a hardware-accelerated processing engine that moves vision AI data processing into programmable network fabric via SmartNICs, freeing CPU resources for inference and mitigating data-pipeline bottlenecks (Glenfinnan, SIGCOMM NAIC '26, first author).
- Co-developed detection-guided attention steering for vision-language models that removes task-irrelevant visual regions before transformer cross-attention, reducing unnecessary model computation (ICML SCALE '26).

Research Intern, Cloud Systems Reliability Group, Microsoft Research *May – Aug 2025*

- Developed a speculative execution stack for LLM web agents that learns reusable website structure to predict data-acquisition strategies before runtime (Skim, EuroSys '27, first author).
- Cut web-agent inference costs by **1.9×** and task latency by **33.4%** on web automation benchmarks while preserving target accuracy.

Research Intern, Networking Research Group, Microsoft Research *Jun – Aug 2022*

- Developed a low-cost datacenter anomaly detector replacing heavy packet traces with lightweight flow summaries, cutting telemetry overhead by **2.7–16.7×** (NetVigil, NSDI '24, co-first author).
- Beat SOTA anomaly-detection AUCs by 0.22 on average; work led to a filed US patent and open-source benchmark and is being explored for deployment within Microsoft's production infrastructure.

Computer Scientist, National Security Agency *Feb 2020 – Aug 2021*

- Maintained and contributed core features to **Ghidra**, the agency's open-source reverse-engineering framework used across global cybersecurity research and defense.
- Built an automated data-augmentation engine applying semantics-preserving binary transformations, improving ML malware-detection AUC by up to **10%** with a **79×** rewrite speedup (ECML PKDD '23, first author).

Research Assistant, Rutgers University / NYU Courant *May 2019 – Jan 2021*
Advisors: Srinivas Narayana, Anirudh Sivaraman

- Co-developed a stochastic program-synthesis eBPF compiler reducing kernel extension sizes by **6–26%** and accelerating equivalence checking by **5 orders of magnitude** (SIGCOMM '21).
- Built a high-fidelity programmable-switch hardware simulator (CoNEXT '20, first author) and co-developed a synthesis-based switch compiler delivering a **51×** average compilation speedup (SIGCOMM '20).

PUBLICATIONS

* denotes equal contribution. Full list at michaeldwong.github.io.

1. **Skim: Speculative Execution for Fast and Efficient Web Agents.** Mike Wong, Kevin Hsieh, Suman Nath, Ravi Netravali. ACM EuroSys 2027 (to appear).
2. **Glenfinnan: SmartNIC-Accelerated Data Processing for Efficient Vision AI Pipelines.** Mike Wong, Ulysses Butler, Emma Farkash, Praveen Tammana, Anirudh Sivaraman, Ravi Netravali. ACM SIGCOMM NAIC 2026.
3. **Detection-guided Attention Steering for Vision Language Models.** Alan Zhang, Rui Pan, Mike Wong, Ravi Netravali. ICML SCALE 2026.
4. **MadEye: Boosting Live Video Analytics Accuracy with Adaptive Camera Configurations.** Mike Wong, Murali Ramanujam, Guha Balakrishnan, Ravi Netravali. USENIX NSDI 2024.
5. **NetVigil: Robust and Low-Cost Anomaly Detection for East-West Data Center Security.** Kevin Hsieh*, Mike Wong*, Santiago Segarra, Sathiya Kumaran Mani, Trevor Eberl, Anatoliy Panasyuk, Ravi Netravali, Ranveer Chandra, Srikanth Kandula (*equal contribution). USENIX NSDI 2024.
6. **Marvolo: Programmatic Data Augmentation for Deep Malware Detection.** Mike Wong, Edward Raff, James Holt, Ravi Netravali. ECML PKDD 2023.
7. **Synthesizing Safe and Efficient Kernel Extensions for Packet Processing.** Qiongwen Xu, Michael D. Wong, Tanvi Wagle, Srinivas Narayana, Anirudh Sivaraman. ACM SIGCOMM 2021. *Also at BPF & Networking Summit, LPC 2021.*
8. **Testing Compilers for Programmable Switches Through Switch Hardware Simulation.** Michael D. Wong, Aatish Kishan Varma, Anirudh Sivaraman. ACM CoNEXT 2020.
9. **Switch Code Generation Using Program Synthesis.** Xiangyu Gao, Taegyun Kim, Michael D. Wong, Divya Raghunathan, Aatish Kishan Varma, Pravein Govindan Kannan, Anirudh Sivaraman, Srinivas Narayana, Aarti Gupta. ACM SIGCOMM 2020.

TECHNICAL SKILLS

Research Areas	AI Systems, Efficient LLM Inference, AI Agents, Computer Vision Systems, Distributed ML Systems, Networked Systems
Languages	Python, C++, C, Rust, Java, JavaScript, Assembly
ML Frameworks	PyTorch, vLLM, HuggingFace
Infrastructure	Docker, Kubernetes, Linux, Git
Networking	eBPF, Verilog, Linux kernel programming, packet processing

PATENTS

Detecting Network Anomalies Using Network Flow Data. Tsuwang Hsieh, Santiago Martin Segarra, Sathiya Kumaran Mani, Srikanth Kandula, Michael Dean Wong. US patent App. 18/367,775.

TEACHING & VOLUNTEERING

Instructor, Prison Teaching Initiative (Princeton University) *Jan 2025 – Dec 2025*
 Volunteer instructor teaching mathematics (MATH 020, MATH 101) to incarcerated students in NJ state correctional facilities.

Teaching Assistant, Princeton University Dept. of Computer Science *Spring – Fall 2023*
 Assisted instruction, built grading infrastructure, held office hours for **COS 316** (Principles of Computer System Design) and **COS 561** (Advanced Computer Networks).